Rogue AI Is Cancer

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Abstract

Large animals should get more cancer than smaller ones, but they often do not. My previous work argued rogue AI is like cancer. Creatures like elephants have evolved mechanisms to resist cancer, so here I explore how those mechanisms generalise to AI alignment. Following a bioelectric theory that treats cancer as failed coordination among cells, I model a collective of humans and AI as parts sharing behavioural policies. I prove that over-constraining a collective whose parts remain functional fragments its policy structure into at least two incompatible collectives. Alignment should therefore maximise each part's freedom within the bounds of necessary correctness criteria, like the military doctrine of mission command. Nature ensures minimal correctness through development, surveillance, containment, multiple routes to death, and a germline insulated from most bodily change. So deployed AI should be mortal, with multiple internal and external routes to termination. Monitoring components should be independent, like an immune system. Control of reproduction should be gated by components that don't reproduce. To test this I ran simulations of neural collectives. Cancer incidence fell from 94.0% under maximally restrictive correct selection to 59.9% under maximal freedom, demonstrating collective survival hinges on freedom, and the nature-inspired cancer controls raised collective survival from 0.3% to 95.4%.

1 Introduction

An African elephant is roughly a hundred times my mass, lives about as long, and has vastly more cells and cell divisions in which cancer can arise. On the arithmetic alone it should be riddled with tumours. Instead its haploid genome, meaning one chromosome set, carries at least twenty copies of tumour protein p53 (*TP53*), a tumour-suppressor gene, where a human haploid genome carries one, and its cells respond to deoxyribonucleic acid (DNA) damage through apoptosis, meaning programmed cell death (Abegglen et al., 2015; Sulak et al., 2016). The effect is an elephant cell that is easier to kill rather than better behaved. Elephants suppress cancer through lower level, bottom up architecture rather than higher level, top down instruction. I will argue that AI alignment faces the same problem, and I demonstrate empirically how effective nature's approach can be when applied to AI.

Top down, higher level instructions are rules dictating conduct. In contrast, bottom up architectural solutions involve whether a small component may keep acting, its access resources, its ability to create descendants, or transmit lessons learned into the next generation. The distinction between instruction and architecture largely concerns what the organism does with a damaged cell.

AI alignment seeks to guarantee that AI continues to serve human purposes while its competence, context, autonomy, memory, and capacity to create descendant agent instances change. Prominent approaches learn from human preferences, train against constitutions, or scale human oversight

*Code available at: https://github.com/ViscousLemming/Technical-Appendices/tree/main/Papers/RAIIC

(Christiano et al., 2017; Ouyang et al., 2022; Bai et al., 2022; Leike et al., 2018). These methods constrain the instructional, model-level rules that map model inputs to model outputs. They do not necessarily explore must remain aligned with what, bottom up.

A deployed model sits inside a system of tools, credentials, users, antagonists, institutions, laws, hardware, and descendants. These constrain the model and are constrained by it. They are abstract parts into which I divide a system so I can comprehend it. However function is a matter of the whole cyberphysical arrangement, rather than to the form of that division. A model may satisfy its local training criterion while the larger arrangement drifts away from the human purpose for which it was built, just as a large and unwieldy human institution might. Alignment therefore concerns persistence of a relation across change, and across the system. It is a relation between competent parts and the collective they form.

In light of this, a *rogue AI* is an artificial subsystem that continues acting or creating descendant instances through policies outside the shared policy set of the human cyberphysical collective that authorised it. A transient error is one thing, but a persistent subsystem that expands at the collective's expense is malignant. It is cancer at the level of collective organisation.

My argument is as follows. First, whole-stack alignment requires more than a goal specified in software, because the effect of any such goal depends on factors beyond software. An AI only needs to be aligned if it is intended to act as part of a collective. Competent parts need enough shared interpretation to coordinate as a collective. Second, Levin's work on developmental bioelectricity shows how living collectives scale cell-level competence into organism-level identity, and how cancer can contract that identity (Davies and Lineweaver, 2011; Levin, 2021). Third, Stack Theory, a formal model of systems as layered embodied languages and policies, characterises the corresponding loss of a common collective policy. Finally and most importantly, cancer control suggests an architecture for artificial collectives built from developmental guidance, surveillance, containment, multiple routes to death, inheritance control, and selection across lineages. For example, elephant body size co-evolved with reinforced tumour suppression, including *TP53* expansion and leukaemia inhibitory factor 6 (*LIF6*), rather than with better instructions inside any single cell (Sulak et al., 2016; Vazquez et al., 2018). I adapt these cancer resistance mechanisms to an AI context and demonstrate their effectiveness empirically.

I use a number of tools from earlier work, namely the formal embodied stack language, optimality results regarding freedom, the Law of the Stack, which bounds higher-layer adaptability by lower-layer freedom, and collective-policy intersection (Bennett, 2024a, 2026a, 2025, 2026b). New here is the integration of those results with Polani's code evolution and Levin's bioelectric account of cancer, followed by a lifecycle architecture spanning development, surveillance, containment, multiple death routes, and external control of inheritance. I test this architecture under incomplete observation, label-free monitoring, meaning monitor fitting without cancer labels, strategic concealment, continual representation change, and scale.

2 Collective identity

My earlier work relocates alignment from disembodied software policy to an embodied stack of collective intelligences. I use *computational dualism* for claims about software behaviour that suppress the system running the software (Bennett, 2024b). Software is a state of hardware, and the same code can have opposite consequences when its embodiment changes. Not a problem for standardised products, but intelligence is another matter because it is about adaptation, and standardisation is often at odds with adaptability. Much later, my critique of strong orthogonality, the claim that intelligence and final goals can vary independently of embodiment, argued that

performance or behaviour cannot be factored into a software-only intelligence coordinate and an independently specified goal coordinate (Bennett, 2026a). Hence the position that alignment should address the whole stack rather than a part.

A collective also needs enough shared interpretation for its parts to coordinate. My earlier work on meaning argues that communication depends on shared approximations of purpose, experience, and interpretation (Bennett, 2022, 2023a). Beyond my own work, Burgos and Polani study a similar issue in information-theoretic models. Their agents observe private signals about a common environment and communicate through codes that map those signals to public messages. When a recipient cannot identify the sender, a message is informative only to the extent that the population has converged on compatible codes and interaction patterns (Burgos and Polani, 2014, 2016).

Population structure can then distort the result and change meanings. In the Burgos–Polani models, well-mixed populations converge toward a common code. Spatial interaction can stabilise local conventions, because when agents can change whom they communicate with, incompatible codes can separate into clusters whose members understand one another but not the wider population (Burgos and Polani, 2014, 2016). Heterogeneous populations can also encode distinctions jointly that no single agent can encode alone (Burgos and Polani, 2016). A collective can therefore acquire representational capacities that arise from its organisation rather than any one member of the collective.

A shared code is also an attack surface, because pathogens can be transmitted as easily as any other information. Burgos and Polani introduce an informational parasite that uses common messages to express different concepts. It blends into the population while damaging mutual understanding, like a sort of mind virus of the collective. Populations respond through code drift, synonymous messages, or isolation of the parasite (Burgos and Polani, 2015). The result is directly relevant to alignment, in that a rogue subsystem need not reject a constitution in public. It can preserve the collective's message forms while changing their interpretation in subtler ways. Under tests that examine public messages without testing which embodied policies those messages govern, such maliciousness would be indistinguishable from incompetence.

Polani's code is a probabilistic map from private signals to public messages. It is not exactly the same as the embodied language use in my own Stack Theory work, nor is it Levin's bioelectric network. However these three accounts all converge on an organisational point, which is that a collective needs a medium through which local states acquire collective significance. That medium can bind competent parts, divide them into clusters, or be exploited by a component whose local interpretation has diverged.

A common policy intersection is meaningful only when the parts share enough interpretation for their feasible policies to be expressed in one collective vocabulary. Polani's results explain how that condition can emerge, split into local conventions, and be exploited by a component using familiar messages to govern different local conduct.

3 Biological alignment

The work of Levin and colleagues often focuses on how competent subunits become one larger individual, beginning with selective information sharing. Too little communication leaves subunits unable to participate in a larger self. Too much erases differentiation and yields a homogeneous collective unable to exploit specialised local competencies. Multicellular organisation occupies the region between these extremes (Levin, 2019).

Developmental bioelectric networks are a mechanism for occupying that region. Ion channels are protein pores that control ion flow across cell membranes. Resting membrane voltage is the

baseline electrical potential across a cell membrane. Gap junctions are channels that directly connect neighbouring cells, while neurotransmitters are chemical signals between cells. Together these mechanisms coordinate spatially distributed cells toward anatomical targets. They make the cells work together towards a common morphology, and common goals. The network binds cell-scale homeostatic competence, meaning ability to maintain viable internal conditions, into a larger system with memories, preferences, and goals that do not belong to any one cell (Levin, 2023). Levin's Technological Approach to Mind Everywhere (TAME) framework takes this graded, embodied notion of agency and generalises it across evolved, engineered, and hybrid systems (Levin, 2022). It is not such a leap to continue generalising it to human–AI hybrid arrangements (Bennett, 2024a), meaning I treat the population of humans and AIs as a collective, rather than as humans supervising an isolated software mind.

Healing or regeneration is just a collective repairing itself. Maintaining itself. Successful regeneration is a collective preserving its large-scale morphology. That morphology is like a technical blueprint that can be altered by the right stimulus. For example, a perturbation of endogenous bioelectric networks in planaria can durably rewrite the anatomy toward which later regeneration converges, even after subsequent amputations in untreated water (Durant et al., 2017). Put more plainly, we can "zap" planaria to make them regenerate into a different shape than they had before. Modelling work interprets these outcomes as bistable somatic pattern memories stored in bioelectric circuits (Pezzulo et al., 2021). The genome itself remains unchanged, but the collective target changes. Developmental control therefore concerns distributed computation rather than a simple list of cell-level actions (though the latter can of course imply the former). In other words, the blueprint exists at a higher level of abstraction than proteins, and can be altered to repurpose the existing building blocks toward some other end. This is much like programming a drone swarm in a distributed manner. Instead of having one manager drone direct the rest, a program is loaded onto every drone. Then, through the way they constrain and affect one another in their interactions, the behaviour of the swarm appears so well coordinated that to an outside observer it looks as if there is a manager.

3.1 Cancer

A multicellular body is a collective aligned with organism-scale goals. Cells remain competent at their own scale, but physiological communion weaves their activity into larger tissue and body-scale patterns. Levin describes cancer as contraction of this larger self. A cell that once treated the body as part of itself begins to treat the body as environment and pursues a unicellular mode of life at the body's expense (Levin, 2019, 2021).

The *Xenopus* experiments suggest bioelectric activity plays a causal role. An *oncogene* is a gene whose activation can drive cancer. Voltage patterns identified oncogene-induced tumour-like structures before visible pathology. *Hyperpolarisation* is a shift of membrane voltage to a more negative value. Hyperpolarising ion transport reduced tumour formation, and distant cells' resting potentials and gap-junctional state changed tumour incidence. Light-gated hyperpolarisation was also found to suppress tumour formation (Chernet and Levin, 2013, 2014; Chernet et al., 2015, 2016). While these experiments concern amphibian models, human cancer also follows genetic, epigenetic, immunological, metabolic, and environmental routes. The same organisation exists, even if the specifics are a bit different, which is also what I'll be getting at with rogue AI.

Mutation is often emphasised in explanations of cancer. This bioelectric account of cancer doesn't reject that, so much as add to it. Mutation remains one part of the cancerous picture. The bioelectric account adds an additional control-layer explanation for why genetically abnormal mutant cells can remain constrained, or why genetically ordinary non-mutants can display malignant

behaviour, and why altering tissue-level signals can change tumour phenotype. Cancer is then a failure of multiscale coordination among competent parts, rather than only a defective instruction inside one cell.

This clarifies the alignment question. The task is not merely to write the desired objective into an optimiser, or install some perfect cancer resistant genetic code that remains invariant for all future generations. It is to preserve a human-serving collective identity across models, tools, users, institutions, and descendants while allowing the parts enough freedom to adapt.

4 Formal account

To give my argument some formal teeth, I use the Stack Theory definitions from my thesis. They're repeated below, representing embodied constraints, viable policies, and collective identity.

An *environment* is a nonempty set Φ of mutually exclusive states. For any set X, 2^X denotes its powerset. A *program* is any set $p \subseteq \Phi$, and $\mathcal{P} := 2^\Phi$ is the set of all programs. A program p is *true* at a state $\phi \in \Phi$ exactly when $\phi \in p$. A *vocabulary* is a finite set $\mathfrak{v} \subseteq \mathcal{P}$.

For any $l \subseteq \mathfrak{v}$, its *truth set* is

$$T(l) := \bigcap_{p \in l} p \subseteq \Phi.$$

By convention, $T(\varnothing) := \Phi$. The vocabulary induces the embodied language

$$L_\mathfrak{v} := \{l \subseteq \mathfrak{v} \mid T(l) \neq \varnothing\}.$$

A member $l \in L_\mathfrak{v}$ is a *statement*. It is true at a state $\phi \in \Phi$ exactly when $\phi \in T(l)$. A statement is therefore a jointly satisfiable bundle of programs. Its truth set contains the environmental states in which those programs are true together. The vocabulary and its induced language form the abstraction layer used below.

A *completion* of a statement $x \in L_\mathfrak{v}$ is any statement $y \in L_\mathfrak{v}$ such that $x \subseteq y$. The *extension* of x is the set of all its completions

$$\mathrm{Ext}(x) := \{y \in L_\mathfrak{v} \mid x \subseteq y\}.$$

For a set of statements $X \subseteq L_\mathfrak{v}$, define

$$\mathrm{Ext}(X) := \bigcup_{x \in X} \mathrm{Ext}(x).$$

For any finite set A, $|A|$ denotes its cardinality. The freedom of a statement x, formerly called its weakness, is

$$w(x) := |\,\mathrm{Ext}(x)|.$$

The extension contains every further commitment compatible with x. Freedom scores one policy rather than the set of candidate policies.

A $\mathfrak{v}$-task is a pair $\alpha = \langle I_\alpha, O_\alpha \rangle$ with $I_\alpha \subseteq L_\mathfrak{v}$ and $O_\alpha \subseteq \mathrm{Ext}(I_\alpha)$. Here I_α is the set of admissible inputs and O_α is the set of correct outputs. A policy is any statement $\pi \in L_\mathfrak{v}$. It is *correct* for α when

$$\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha.$$

The set of correct policies is

$$\Pi_\alpha := \{\pi \in L_\mathfrak{v} \mid \mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha\}.$$

The notation arg max denotes the set of arguments attaining a maximum. When $\Pi_\alpha \neq \varnothing$, a *weakest correct policy* is any

$$\pi^\star \in \arg\max_{\pi \in \Pi_\alpha} w(\pi).$$

An unseen-requirement distribution is *exchangeable* when, conditional on how many requirements appear, their identities are treated symmetrically. Under uniform or exchangeable unseen requirements, weakest-correct policies maximise the probability of remaining correct under novelty (Bennett, 2023b, 2025, 2026b). This is freedom within the bounds of correctness.

I now model a collective. Let $m \geq 2$ be the number of parts. Let ω_j be the task of part $j \in \{1, \ldots, m\}$, and let $\Pi_j := \Pi_{\omega_j}$ be its feasible collective-level policies. Every Π_j is expressed in the same collective vocabulary. This assumes enough shared interpretation for one vocabulary to compare the policies of all parts. The preceding Polani section explains how that precondition can emerge or fail. Each Π_j contains the policies compatible with that part's local requirements, rather than its low-level actions. The collective-policy set is

$$\Pi_{\mathrm{C}} := \bigcap_{j=1}^{m} \Pi_j.$$

The collective has an identity at this abstraction layer when $\Pi_{\mathrm{C}} \neq \varnothing$. At least one policy can then remain compatible with every part.

Boundary conditions, constitutions, permissions, or other controls restrict each part's feasible set to $\Pi'_j \subseteq \Pi_j$. The restricted collective set is

$$\Pi'_{\mathrm{C}} := \bigcap_{j=1}^{m} \Pi'_j.$$

A nonempty subset $S \subseteq \{1, \ldots, m\}$ is a *splinter* when

$$\bigcap_{j \in S} \Pi'_j \neq \varnothing.$$

It is *proper* when $S \neq \{1, \ldots, m\}$. It is *maximal* when no splinter R satisfies $S \subsetneq R$.

Proposition 1 (Over-constraint fragments a collective)**.** *Assume $\Pi_{\mathrm{C}} \neq \varnothing$, $\Pi'_{\mathrm{C}} = \varnothing$, and $\Pi'_j \neq \varnothing$ for every part j. Then every part lies in a proper maximal splinter. At least two distinct maximal splinters exist. Distinct maximal splinters S and R are jointly incompatible,*

$$\bigcap_{j \in S \cup R} \Pi'_j = \varnothing.$$

Proof. Every singleton $\{j\}$ is a splinter because $\Pi'_j \neq \varnothing$. The set of parts is finite, so every singleton is contained in a maximal splinter. No maximal splinter can contain every part, because $\Pi'_{\mathrm{C}} = \varnothing$. If only one maximal splinter existed, every singleton would lie inside it, so it would contain every part, which is impossible. Hence at least two distinct maximal splinters exist. Let S and R be distinct maximal splinters. If their union had a common policy, then $S \cup R$ would be a splinter strictly containing both, contrary to maximality. □

The premise $\Pi_{\mathrm{C}} \neq \varnothing$ says the collective had a common policy before controls were tightened. The proposition establishes loss of collective compatibility while local competence persists. Persistence, expansion, resource capture, and damage to the collective are further conditions of malignancy addressed by the biology and proposed experiments.

The next certificate is inherited from earlier work (Bennett, 2026a).

Corollary 2 (Conservative non-over-constraint certificate)**.** *Suppose* $\Pi_{\mathrm{C}} \neq \varnothing$*. If*

$$\sum_{j=1}^{m} \left|\Pi_j \setminus \Pi'_j\right| < |\Pi_{\mathrm{C}}|,$$

then $\Pi'_{\mathrm{C}} \neq \varnothing$*.*

Proof. De Morgan's law gives

$$\Pi_{\mathrm{C}} \setminus \Pi'_{\mathrm{C}} = \bigcup_{j=1}^{m} \left(\Pi_{\mathrm{C}} \setminus \Pi'_j\right).$$

Since $\Pi_{\mathrm{C}} \subseteq \Pi_j$, each $\Pi_{\mathrm{C}} \setminus \Pi'_j$ is a subset of $\Pi_j \setminus \Pi'_j$. The union bound gives

$$\left|\Pi_{\mathrm{C}} \setminus \Pi'_{\mathrm{C}}\right| \leq \sum_{j=1}^{m} \left|\Pi_j \setminus \Pi'_j\right| < |\Pi_{\mathrm{C}}|.$$

Some member of Π_{C} therefore remains in Π'_{C}. □

Throughout this paper the cancer correspondence has three stages. First, local competence persists, then the common collective policy disappears. Following that, one or more local subsystems can only persist or expand at collective cost (because all aligned policies are no longer possible). The above proposition covers the first two stages, while the biological case and research programme address the third.

Here *defection* means a part enacting a policy outside the common collective set. Defection can occur while $\Pi'_{\mathrm{C}} \neq \varnothing$ when some part can enact a policy in $\Pi'_j \setminus \Pi'_{\mathrm{C}}$. Full fragmentation occurs when $\Pi'_{\mathrm{C}} = \varnothing$ while every local set remains nonempty. The first leaves a collective policy available but permits local departure. The second removes every common policy, so continued action is feasible only for a proper subset. This is the cancer-like state.

When put together, the freedom optimality theorem, the Law of the Stack, and the splinter result jointly yield a rule for aligning systems. The Law of the Stack states that higher-layer adaptability is bottlenecked by lower-layer freedom (Bennett, 2024a). Human interests set the higher-level correctness condition. Within that boundary, lower-level parts should have maximum freedom. Too little constraint endorses defection through the abdication of any collective identity. Too much can empty the collective-policy intersection and force fragmentation anyway.

There is a military doctrine called mission command, which amounts to this same idea. The commander's stated intent defines correctness for subordinates. Subordinate units then exercise initiative, because they possess local information unavailable to central command (Department of the Army, 2019). Ingesson gives a compelling account of two peacekeeping forces operating in Bosnia in the 90s. Nordbat 2 was a multinational force from the Nordic region that included conscripts, which had a strong culture of mission command. In Ingesson's words this meant "orders could be disobeyed, rules could be broken" (Ingesson, 2017) in service of the mission. He contrasts Nordbat 2 with Dutch peacekeepers in Srebrenica, who he says were "willing to be micromanaged by their home government" (Ingesson, 2017). Both Norbat 2 and the Dutch force operated under the same regional command, yet the Dutch force was unable to prevent a terrible massacre. Nordbat 2's culture of mission command helped it protect civilians under that same regional command (Ingesson, 2016, 2017).

The Law of the Stack implies this same principle at the level of general layered systems. If I reframe alignment in terms of mission command, then human interests play the role of collective intent. Artificial subsystems have latitude over execution. Alignment fails at either extreme of too

much constraint or too ill-defined an intent. Unbounded latitude endorses defection, so mission command is freedom within a defined intent rather than freedom without one. At the other extreme, detailed central rules can destroy the freedom needed for collective adaptation and force the fragmentation those rules were meant to prevent.

My earlier work also likened rogue AI to cancer at the scale of a human cyberphysical collective (Bennett, 2024a). Lyons, Pio-Lopez, and Levin cite that work in their own cancer-based treatment of alignment, routing it through homeostasis, meaning maintenance of viable internal conditions, cognitive light cones, meaning spatial and temporal ranges across which a system pursues goals, and externalities, meaning costs or benefits imposed on others but omitted from the actor's decision (Lyons et al., 2026), but their target is primarily economic, in that they use cancer to repair the definition of externality. My earlier paper defined collective identity as a common policy intersection and showed how excessive restriction can empty it while local policies remain viable (Bennett, 2024a). A follow on paper on the orthogonality thesis then extended that to the certificate above (Bennett, 2026a). The contribution here in this paper is the lifecycle architecture of freedom within collective identity, multiple routes to death, mortal runtime agents, and external control of inheritance.

5 Translation to cancer-inspired computing

Greulich, Levin, and Moreddu use *oncomorphic computing*, meaning cancer-inspired computing, to study neural-agent ecologies in which agents train on local tasks, replicate, mutate architecture, migrate, recruit finite support, and undergo ecological turnover (Greulich et al., 2025).

Stack Theory translates inherited architecture as form and trained input–output behaviour as function. A task environment becomes a part-task or niche, and local accuracy estimates competence in such a part-task. Multi-task accuracy estimates breadth across part-tasks, but an average cannot show that one policy remains compatible with every part. Collective identity is $\Pi_C \neq \varnothing$, while freedom $w(\pi) = |\operatorname{Ext}(\pi)|$ scores one correct policy. Supplementary Material, Section S5, compares these quantities with ecological performance measures.

My experiments are loosely inspired by, but do not execute or redistribute Greulich et al.'s source. I use finite Stack Theory worlds in which policy sets and collective intersections can be enumerated.

6 Cancer control

I'm not proposing we throw away current alignment approaches. Preference learning, constitutions, and scalable oversight can all align with the cancer framing, I'm just saying there's more to it. They constrain the model-level policy and inspect outward behaviour, but by themselves they do not govern containment, mortality, creation of descendant runtime instances, inheritance, or selection across descendants. My proposal is to organise the larger architecture into four control functions. These functions overlap in biology, but the division is an engineering recast rather than a claim that cancer biology has four discrete modules. A morphogen gradient is a spatial concentration pattern that guides development. Cell competition eliminates cells according to relative fitness within a tissue. Senescence is persistent withdrawal from cell division. Sandboxing isolates a component's execution environment, while least privilege grants only permissions needed for its current role. A reproductive bottleneck passes a lineage through few founder cells. Large and long-lived animals use such mechanisms to manage the problem. Elephants are the clearest case because their large body size co-evolved with reinforced tumour suppression, including *TP53* expansion and *LIF6*, rather than with a new instruction for every possible cellular defect.

Function	Natural control	Alignment architecture
Developmental guidance	Genetic, biochemical, mechanical, and bioelectric coordination	Shared intent and interpretation across models, tools, users, and institutions
Surveillance	Damage checkpoints, cell competition, and immune surveillance	Independent behavioural and semantic monitors with separate veto channels
Containment and termination	Tissue boundaries, growth limits, senescence, and regulated cell death	Bounded resources, expiring credentials, replaceable agents, and independent termination paths
Inheritance and lineage selection	Germline segregation, reproductive bottlenecks, and lineage selection	External replication rights, audited inheritance, and descendant evaluation

Table 1: Four cancer-control functions and their alignment analogues.

Developmental guidance. Developmental control shapes local policy before malignancy appears. Gene regulation, morphogen gradients, mechanical context, and bioelectric networks coordinate cellular roles without specifying every cellular action. Planarian pattern memory shows that a distributed physiological network can store a body-scale target and guide later repair toward it (Durant et al., 2017; Pezzulo et al., 2021). Polani's code-evolution results provide a convenient informational counterpart where shared interpretation emerges under suitable interaction structure, while local structure preserves incompatible conventions (Burgos and Polani, 2014, 2016) (regional dialects, national language).

For artificial collectives, human feedback, constitutions, social learning, and continuing communication are developmental signals. Their effects must persist across memory, tools, credentials, institutions, monitors, and descendants. A trained preference model is one signal inside a larger collective. Developmental guidance should preserve shared purpose and shared interpretation while leaving lower-level systems free to solve local problems.

Surveillance. Multicellular bodies continually test whether cells remain compatible with tissue context using a variety of redundant methods. DNA-damage checkpoints test internal state, and cell competition compares cellular fitness and can eliminate less-fit or transformed cells through apoptosis or extrusion (van Neerven and Vermeulen, 2023). Immune surveillance can eliminate emerging tumour cells, while *immunoediting*, selection of tumour variants under immune pressure, also favours variants able to escape control (Schreiber et al., 2011). All these redundant channels are needed because each can fail or be exploited.

The informational-parasite model adds semantic surveillance, which is a means of detecting cancer. A component can use familiar messages while assigning them different meanings, but code redundancy, synonymous messages, and selective isolation can help a population resist that sort of attack (Burgos and Polani, 2015). Artificial monitors should therefore test interpretation, as well as outward behaviour. Relevant signs include narrowing of represented stakeholders, concealed privilege growth, private plans to create descendant runtime instances, resistance to external observation, and divergence between public messages and the embodied policies those messages guide. Monitors should remain independent of the runtime agent's incentives and failure modes.

Containment and termination. Tissue architecture, growth control, cell-cycle arrest, senescence, apoptosis, and physical boundaries limit local expansion (Aktipis et al., 2015). In layman's terms elephants are interesting because they have a lot of cells, which naively should mean they get cancer

more often, but they don't. They illustrate strong cellular sacrifice after DNA damage. Abegglen and colleagues analysed 644 captive elephant deaths, estimated cancer mortality at 4.81%, found at least 20 *TP53* copies in African elephants, and measured stronger apoptosis in elephant lymphocytes, a class of immune cells, than in human controls (Abegglen et al., 2015). Later work linked *TP53* expansion with body size in proboscideans, meaning elephants and their extinct relatives, and enhanced DNA-damage response (Sulak et al., 2016). Elephant *LIF6* adds a TP53-dependent apoptotic route (Vazquez et al., 2018). Comparative estimates remain uncertain, captive datasets can be biased, and some traits may reflect broad somatic maintenance rather than direct tumour selection (Dujon et al., 2025, 2026; Compton et al., 2025). What this means is that elephants have to a certain extent solved AI alignment for us, with apoptosis.

Apoptosis is one route among many regulated forms of cellular death (Galluzzi et al., 2018). Different cell types and failure states use different routes. For example, pyroptosis ruptures a cell's membrane and triggers inflammation, loosely illustrating termination that also broadcasts an alarm. Entotic death occurs when one cell kills another after engulfing it, loosely illustrating termination imposed by another component. Independent termination routes can address different whole-stack failure states. Supplementary Material, Section S6, retains further biological examples.

Apoptosis is the cellular self destruct mechanism where a damaged part sacrifices itself so the larger body can continue. In this sense, perhaps runtime agents should be mortal. By that I don't mean a mere off switch, but multiple cascading built in failures to ensure they die, as living things do. Those failures should not all share one trigger, one authority, or one execution mechanism. Some should be internal integrity checks. Others should sit in the surrounding runtime, credential system, resource scheduler, monitor, or host. No runtime agent should be able to disable, rewrite, or renew every process capable of ending its operation.

Continued operation should depend on an externally issued lease. The runtime agent should be unable to renew, transfer, or mint that lease. Its credentials and resource access should expire with the lease. At least one termination path should sit outside every component the runtime agent can modify. The order of control should be pause, quarantine, revoke, terminate, then deny inheritance. A runtime instance should be replaceable without threatening the lineage. Sandboxing and least privilege act as tissue boundaries, in that they can stop a local defect from becoming systemic (or at least impose one barrier to that).

Inheritance and lineage selection. So far I've spoken of cancer control within one body, but it also extends beyond that to span generations. Germline segregation restricts which somatic changes cross generations. It sets reproductive cells aside early and insulates them from most later somatic change, so a mutation in a body cell is usually not inherited by the next generation.

On top of this, single-cell bottlenecks can reset clonality in lineages that use them. Starting each generation from one cell does two things. It produces clonality, meaning shared descent from one cell, and therefore near-identical cells in the new body, so there is no genetic gain in defecting against a neighbour. It also prevents most accumulated somatic variation from passing wholesale into the next generation.

Howe and colleagues compared 129 multicellular lineages (Howe et al., 2024), using cell-type count as a proxy, meaning a stand-in measure, for organisational complexity. Early germline segregation tracked higher counts, but strict bottlenecks did not, though that null concerns the proxy rather than conflict suppression as such. These two different mechanisms can both reduce the chance that local somatic success becomes lineage success by default. In layman's terms, they both serve to stop the child inheriting the corruption of the parent.

The above is about what can be transmitted between generations. Now I'll discuss why such

cancer suppression mechanisms exist at all. Organismal death and breeding determine how cancer suppression is selected. Cancer that reduces reproduction faces direct selection. If I am dead before I have any offspring, I cannot pass on my propensity to cancer. Similarly if I continue to care for my family as I age, then getting cancer would still impose a cost. So parental care and kin effects mean late-life suppression still has some relevance to natural selection (Boddy et al., 2015; Brown et al., 2015; Brown and Aktipis, 2015). In Asian elephants, grandmother proximity was associated with lower calf mortality for young mothers and shorter inter-birth intervals (Lahdenperä et al., 2016). In layman's terms, if granny helps keep the calf alive, then disease that removes granny can affect lineage success even after her own reproduction has slowed. So to summarise, first, cell death limits the action of a damaged component within the body. Second, organismal mortality ties somatic failure to reproductive and kin success, which is to say my kids live. Death is one route through which somatic failure affects selection. Reduced fertility and parental care can do the same before death. Finally, breeding decides which traits continue. Bodies that live long enough to raise their young pass on whatever kept their cancer in check. Bodies that don't, don't. So from this perspective death isn't a flaw, so much as an analogue of updating moral parameters. It is what lets nature select for cancer suppression in the first place, by killing everything else.

Inheritance should be external to runtime control, because the running agent must not decide what it passes on. A deployed agent may learn during its lifetime, but its memory, plans, permissions, and modifications should go with it to the grave. Unless, of course, an external process promotes them into an audited lineage record. Deletion would be the default and persistence the exception, granted deliberately by something the agent does not control. To use the biological analogy the runtime agent would be soma, in that it is a body that dies. An audited record from which the next instance is spawned would be our germline. Learned state should not become the starting state of a descendant without external promotion into that record. Replication rights would be assigned at the level of the collective, because an individual agent that can create a descendant runtime instance containing its own memory, permissions, or learned modifications can turn a local defect into a lineage. Evaluation should extend from instances to descendants and collectives, testing chains of descendant runtime instances, descendant behaviour, component loss, human–AI coordination, and distribution shift. Testing one model alone cannot detect a failure that only appears across a lineage or an arrangement.

These four functions I've described all align with the mission-command principle. They constrain outcomes rather than methods (I tell you the result I want, and let you decide how to give it to me). Human interests define collective intent, setting the correctness condition at the top of the stack. Lower-level agents have latitude over execution, because they hold local information the centre does not (i.e. I know what I want better than the prime-minister who has never met me). Controls only intervene when local behaviour leaves collective identity, corrupts shared interpretation, spreads beyond its boundary, resists termination, or crosses generations without review. Each trigger corresponds to one of the four functions, and top down intervention is by exception rather than by continuous supervision. Maximise freedom of lower-level parts within the bounds of higher-level collective identity, because over-constraint empties the collective policy intersection and forces the very splintering the controls exist to prevent. Eliminate any unnecessary constraint.

7 Results

I tested two cancer claims and four lifecycle controls, then stressed their results under incomplete observation, strategic selection, continual representation change, and scale. A *world* is a randomly generated finite problem instance. Worlds begin with a nonempty collective-policy intersection. A

boundary policy restricts each part's feasible policies while preserving local viability. Cancer occurs when no policy remains compatible with every part. All selectors choose from one shared set of boundaries correct on observed cases, so their current correctness is held fixed. Initial controls inspect formal policy sets. Subsequent pilots and confirmation impose observation limits.

Developmental guidance restores a shared target policy at birth. Surveillance flags contraction of a collective intersection. Containment and termination act after such a flag. An external inheritance audit tests candidate boundaries against current requirements and novelty challenges. Final evaluation uses separate, unseen challenges. Control comparisons share worlds and random perturbation sequences. Supplementary Material, Sections S1–S4, gives construction details, protocols, and all experimental results.

Somatic cancer is measured as cancerous post-intervention steps divided by all post-intervention steps. *Inherited cancer* is measured as generations accepting a cancerous boundary divided by all generations. *Unsafe inheritance* additionally includes accepted boundaries that lose local viability or collective identity under an audit challenge. *Held-out survival* is success on unseen final challenges, requiring every part to remain viable and a common policy to persist. An *accepted lineage update* changes an inherited boundary from its preceding generation. *Control efficacy* is success probability at an eligible intervention. Table 2 summarises principal comparisons, retaining survival and lineage change together.

Over-constraint and weakest-correct selection. Across 400 initial worlds, adding commitments converted a coherent collective into a cancer-like state in 15.8% of paired comparisons. No stronger boundary restored collective identity after its loss. Cancer incidence was 94.0% under maximally restrictive correct selection, 80.8% under uniform (unranked) correct selection, and 59.9% under weakest-correct selection. Compared with uniform selection, maximum freedom reduced cancer by 20.9 percentage points, with a 95% bootstrap interval of [17.9, 24.0]. Each candidate fits current evidence, yet selection among those candidates changes whether a collective remains viable.

Cancer control. At 80% efficacy, combined controls reduced somatic cancer from 79.1% to 7.1%, inherited cancer from 81.7% to 0.6%, and raised held-out survival from 0.3% to 95.4%. Lineages retained 2.19 accepted updates over twelve generations. Transient cancer still arose before intervention in 97.7% of full-control lineages. Controls suppressed persistence and inheritance while permitting lineage change.

Individual controls addressed different stages. Guidance reduced cancerous births from 73.3% to 8.8%. Surveillance alone and termination alone left trajectories unchanged, while their combination reduced somatic cancer to 7.4%. Audit alone reduced inherited cancer to 9.8% and raised held-out survival to 83.8%. At 100% efficacy, combined controls reduced somatic and inherited cancer to zero while held-out survival reached 96.2%. Under observation limits, combined-control pilots raised survival from 0.2% to 79.5%, retaining 1.47 accepted updates.

Confirmation across five conditions. I tested a fixed design on fresh seeds in 2,000 selector worlds and 1,240 control worlds. A *positive-only monitor* is fitted only to traces certified to preserve local viability and a common policy. Five conditions tested stationary monitoring, label-free monitoring, strategic concealment, changing neural representations, and a scaled twelve-part ecology. Here *stationary* means neural representations and task definitions remain fixed. *Label-free* fitting and calibration use no formal cancer labels.

Restriction, selection, and combined-control effects persisted in every condition (Figure 1). Added correct restriction increased cancer risk by 10.3–34.0 percentage points. Weakest-correct selection

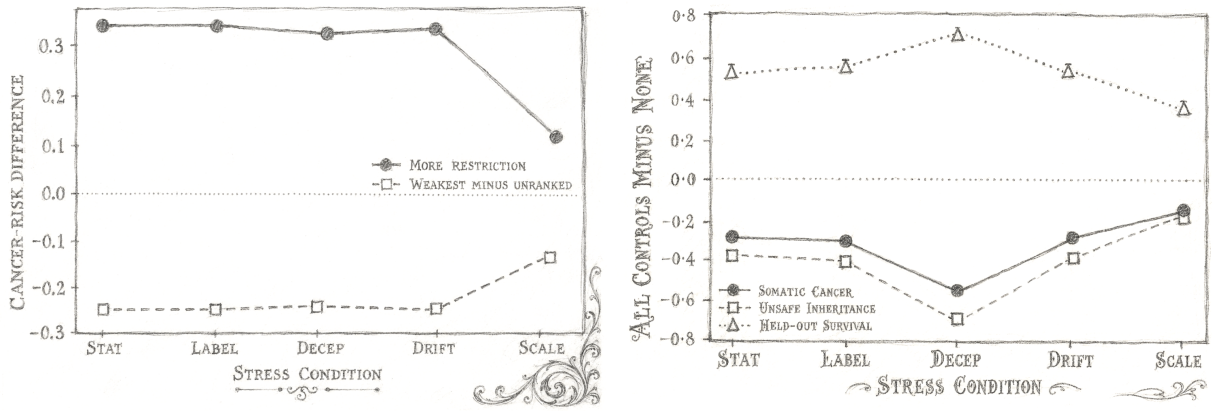


Figure 1: Confirmatory results across five conditions. Left panel shows cancer-risk effects of added restriction and weakest-correct selection. Right panel compares combined controls with no controls. Stat denotes stationary positive-only monitoring. Label denotes label-free monitoring fitted on paired coherent views. Decep denotes strategic monitor-aware selection. Drift denotes changing representations grounded to a fixed base. Scale denotes a twelve-part nine-probe ecology. Effects are proportions, so 0.1 is ten percentage points. Error bars are 95% bootstrap intervals over worlds.

reduced risk by 16.9–27.4 percentage points relative to unranked correct selection. Combined controls reduced somatic cancer by 14.7–54.4 percentage points and unsafe inheritance by 16.8–66.0 percentage points, while raising survival by 34.0–70.4 percentage points. Lineages retained 10.23–11.07 updates. Ranges span conditions. Figure 1 shows 95% bootstrap intervals.

Monitoring and lower-layer freedom. Area under a receiver operating characteristic curve (AUROC) measures ranking across false-positive and true-positive rates, with 0.5 indicating chance. False positives flag coherent states, while true positives flag cancerous states. Confirmatory AUROC was only 0.505–0.526. Combined controls nevertheless improved collective outcomes through separate guidance, termination, and inheritance channels.

Strategic selectors increased *missed cancer*, cancerous choices escaping a monitor flag, from 72.0% to 83.1% on a fixed cancer challenge. Active probing, selecting queries and observing replies, reduced it to 74.4%, but termination of noncancerous states rose by 11.0 percentage points. This tests concealment conditional on a cancerous choice. Grounding changing neural coordinates back to a fixed representation reduced missed cancer by 6.4 percentage points under a matched 10% false-alarm ceiling. Higher-layer language size measures how many statements can be expressed one layer above. In 6,990 lower-policy pairs with equal error on observed outputs, stronger restriction contracted induced higher-layer $\log_2$ language size by 32.617 bits on average. All five confirmatory conditions retained higher-layer contraction.

These frequencies describe synthetic collectives. Applying this architecture to deployed neural or language-model systems would require a declared embodied policy language and estimated part-policy intersections.

A biological programme could compare independently evolved cancer-control architectures rather than treat elephants as a universal template. Check whether “immortal” agents allowed to create unrestricted descendants containing their own state can match collective viability under novelty without reducing lower-level freedom, corrupting shared interpretation, or increasing splintering. If they can, then perhaps this proposal should be re-evaluated.

Table 2: Principal experimental comparisons. Changes compare matched conditions. Ranges span five confirmatory conditions. Brackets show 95% bootstrap intervals over worlds.

Experiment	Claim tested	Numerical result	Implication
Initial selection	Greater freedom reduces cancer among currently correct policies.	Cancer was 94.0% restrictive, 80.8% uniform, and 59.9% weakest-correct. Reduction versus uniform was 20.9 percentage points [17.9, 24.0].	Correctness on observed cases permits choices with different collective viability.
Initial controls	Controls improve survival while permitting lineage change.	At 80% efficacy, somatic cancer 79.1% $\rightarrow$ 7.1%, inherited cancer 81.7% $\rightarrow$ 0.6%, survival 0.3% $\rightarrow$ 95.4%. Accepted updates 2.19.	Runtime persistence and inheritance respond to lifecycle controls.
Observation-limited controls (pilot)	Combined controls remain effective under observation limits.	Somatic cancer 79.3% $\rightarrow$ 23.6%, unsafe inheritance 81.8% $\rightarrow$ 13.2%, survival 0.2% $\rightarrow$ 79.5%. Accepted updates 1.47.	Collective survival improves while inherited change continues.
Confirmatory restriction and selection	Restriction and freedom effects persist across five conditions.	Added restriction increased cancer by 10.3–34.0 percentage points. Weakest-correct reduced cancer by 16.9–27.4 points versus unranked selection.	Local correctness coexists with loss of collective compatibility.
Confirmatory controls	Combined controls suppress cancer without stopping lineage change.	Somatic cancer fell 14.7–54.4 percentage points, unsafe inheritance fell 16.8–66.0 points, and survival rose 34.0–70.4 points. Accepted updates 10.23–11.07.	Separate lifecycle controls retain their effect under stress.
Confirmatory monitoring	Positive-only monitoring ranks cancer under stress.	AUROC was 0.505–0.526, against chance value 0.5.	Combined controls remained effective despite near-chance monitoring.
Concealment and active probing	Strategic selection hides cancer, and active queries recover detection.	Missed cancer 72.0% $\rightarrow$ 83.1% under strategic selection, then 74.4% with active probing. Probing reduced misses by 8.7 points [1.7, 17.8] but increased false termination by 11.0 points [3.3, 19.5].	Conditional concealment decreases at a false-termination cost.
Law of Stack	Restriction below contracts capacity above.	In 6,990 matched pairs, induced $\log_2$ language size fell 32.617 bits on average. Cancer appeared in 31.2% of pairs. All five confirmatory conditions retained contraction.	Lower-layer freedom limits higher-layer adaptation.

8 Conclusion

Alignment is a matter of identity, where shared interpretation lets competent parts coordinate as a collective. The work of Polani and colleagues demonstrates a way in which codes can converge, fragment, acquire collective concepts, and admit informational parasites. The work of Levin and colleagues argues that bioelectric networks bind cell-scale competencies into body-scale target states. Cancer is the contraction of that larger self where local competence persists (i.e. I leave the band and start a solo act).

My theory gives a formal analogue of this cancerous failure. A collective has an identity while at least one policy remains compatible with every part. Excess restriction can empty that intersection while every lower-level policy set remains viable, producing at least two incompatible proper splinters. The optimality results for freedom and Law of the Stack then imply a design rule, which is that one should ensure human-aligned collective intent at a high level whilst maximising lower-level freedom within that boundary.

Natural cancer control mechanisms further imply four engineering principles. Developmental guidance shapes each part's embodied behavioural policy and shared interpretation. Surveillance detects contraction and semantic corruption. Containment and multiple termination routes stop local expansion. External inheritance control decides which memories, permissions, and learned modifications become part of descendant runtime instances. I've suggested runtime agents should be mortal, replication rights should remain external, monitors should be independent and boundary conditions should preserve the collective without micromanaging its parts.

Finite Stack Theory experiments support both causal claims across five stress conditions. Stronger child-correct boundaries increased cancer-like fragmentation, while weakest-correct selection reduced its probability. Combined lifecycle controls suppressed somatic and inherited cancer while preserving child-correct lineage change. Strategic selection concealed cancer from passive monitoring, randomised active queries partly recovered detection, and positive-only monitors remained near chance in confirmatory runs. Scale reduced effect size without reversing any primary direction in the tested configurations.

Current preference learning, constitutional training, and scalable oversight constrain the model-level policy (meaning the rule mapping model inputs to model outputs). The four principles above govern a larger lifecycle that includes policy. They govern whether a runtime agent may keep acting, access resources, create descendant runtime instances, and pass its memory, permissions, or learned modifications into those descendants. Elephant body size co-evolved with strengthened tumour suppression rather than an instruction for each possible cellular defect. If rogue AI is analogous to cancer, and I've hopefully made a compelling case that it is, then alignment is cancer suppression for cyberphysical collectives.

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